

Research Notes on Markov decision process

Markov decision process

>> Section 1

Maximize $\sum_{i \in S} \sum_{a \in A(i)} R(i, a) y(i, a)$ s.t. $\sum_{i \in S} \sum_{a \in A(i)} q(j | i, a) y(i, a) = 0 \forall j \in S, \sum_{i \in S} \sum_{a \in A(i)} y(i, a) = 1, y(i, a) \geq 0 \forall a \in A(i)$ and $\forall i \in S$

$$\begin{aligned} & \sum_{i \in S} \sum_{a \in A(i)} R(i, a) y(i, a) \\ & \sum_{i \in S} \sum_{a \in A(i)} q(j | i, a) y(i, a) = 0 \quad \forall j \in S, \\ & \sum_{i \in S} \sum_{a \in A(i)} y(i, a) = 1, \quad y(i, a) \geq 0 \quad \forall a \in A(i) \text{ and } \forall i \in S \end{aligned}$$

>> Section 2

$R_a(s, s')$ is the immediate reward (or expected immediate reward) received after action a is taken to transition from state s to state s' . The reward is in general a random variable. A policy function π is a (potentially probabilistic) mapping from state space (S) to action space (A) .

>> Section 3

A Markov decision process (MDP) is a mathematical model for sequential decision making when outcomes are uncertain. It is a type of stochastic decision process, and is often solved using the methods of stochastic dynamic programming. Originating from operations research in the 1950s, MDPs have since gained recognition in a variety of fields, including ecology, economics, healthcare, telecommunications and reinforcement learning. Reinforcement learning utilizes the MDP framework to model the interaction between a learning agent and its environment. In this framework, the interaction is characterized by states, actions, and rewards. The MDP framework is designed to provide a simplified representation of key elements of artificial intelligence challenges. This modeling framework incorporates the understanding of cause and effect, the management of uncertainty and nondeterminism, and the pursuit of explicit goals. The name comes from its connection to Markov chains, a concept developed by the Russian mathematician Andrey Markov. The "Markov" in "Markov decision process" refers to the underlying structure of state transitions that still follow the Markov property. The process is called a "decision process" because it involves making decisions that influence these state transitions, extending the concept of a Markov chain into the realm of decision-making under uncertainty.

>> Section 4

$\sum_{i \in S} \sum_{a \in A(i)} R(i, a) y^*(i, a) \geq \sum_{i \in S} \sum_{a \in A(i)} R(i, a) y(i, a)$

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>> Section 5

$V(s) := \sum_{s'} P_{\pi}(s, s') (R_{\pi}(s, s') + \gamma V(s'))$

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>> Section 6

Continuous-time Markov decision process In discrete-time Markov Decision Processes, decisions are made at discrete time intervals. However, for continuous-time Markov decision processes, decisions can be made at any time the decision maker chooses. In comparison to discrete-time Markov decision processes, continuous-time Markov decision processes can better model the decision-making process for a system that has continuous dynamics, i.e., the system dynamics are defined by ordinary differential equations (ODEs). This modelling framework can be applied to areas such as queueing systems, epidemic processes, and population processes. Like the discrete-time Markov decision processes, in continuous-time Markov decision processes the agent aims to find the optimal policy that would maximize the expected cumulative reward. The key difference with the standard case is that, due to the continuous nature of the time variable, summation is replaced by an integral:

>> Section 7

Minimize g s.t. $g - \sum_{j \in S} q(j | i, a) h(j) \geq R(i, a) \forall i \in S, a \in A(i)$

$$\begin{aligned} & \text{Minimize } g \\ & g - \sum_{j \in S} q(j | i, a) h(j) \geq R(i, a) \quad \forall i \in S, a \in A(i) \end{aligned}$$

>> Section 8

$P_a(s, s')$ is the transition of the system, which in this case is going to be deterministic and driven by the laws of mechanics.

>> Section 9

Further reading Bellman, R. E. (2003) [1957]. Dynamic Programming (Dover paperback ed.). Princeton, NJ: Princeton University Press. ISBN 978-0-486-42809-3. Bertsekas, D. (1995). Dynamic Programming and Optimal Control. Vol. 2. MA: Athena. Derman, C. (1970). Finite state Markovian decision processes. Academic Press. Feinberg, E.A.; Shwartz, A., eds. (2002). Handbook of Markov Decision Processes. Boston, MA: Kluwer. ISBN 9781461508052. Guo, X.; Hernández-Lerma, O. (2009). Continuous-Time Markov Decision Processes. Stochastic Modelling and Applied Probability. Springer. ISBN 9783642025464. Meyn, S. P. (2007). Control Techniques

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>> Section 10

$$\max_{a(t) = \pi(s(t))} \int_0^T \text{Tr}(s(t), a(t)) dt + D[s(T)]$$
$$\text{s.t. } ds(t) = f[t, s(t), a(t)] dt + D[s(T)] \quad \text{with } s(0) = s_0$$

>> Section 11

$P_a(s, s')$ is, on an intuitive level, the probability that action a in state s at time t will lead to state s' at time $t + 1$. In general, this probability transition is defined to satisfy $\Pr(s_{t+1} \in S' \mid s_t = s, a_t = a) = \int_{S'} P_a(s, s') ds'$, for every $S' \subseteq S$ measurable. In case the state space is discrete, the integral is intended with respect to the counting measure, so that the latter simplifies as $P_a(s, s') = \Pr(s_{t+1} = s' \mid s_t = s, a_t = a)$; in case $S \subseteq \mathbb{R}^d$, the integral is usually intended with respect to the Lebesgue measure.